

Star schema. Four-step process

Step 1 — Select the business process

From the business perspective the Sales Transactions is the core business process to model. The products are sold to customers.

Step 2 — Declare the grain

One row represents a single product item sold within a transaction

Step 3 — Identify the dimensions

- CUSTOMER (customer + address chain)
- EMPLOYEE (employee + address chain)
- STORE (store + address chain)
- PRODUCT (product + its category)
- DATE

Step 4 — Identify the facts

SALES

I think it will be reasonable to combine PAYMENTS and SALES table into single fact table. SALES has no numeric measure but only foreign keys and a date. Only PAYMENTS table has measure(amount) and it contains sale_id attribute, so the assumption is that one payment is related to one sale. It is helpful that PAYMENTS does not contain product_id, so assumption that amount is the amount paid for specific product item, not order total amount.

Snowflake schema. Normalization steps

- Product Chain: DIM_PRODUCT > DIM_CATEGORIES
- Address Chain: DIM_CUSTOMER / DIM_STORE / DIM_EMPLOYEE > DIM_ADDRESS > DIM_CITY > DIM_COUNTRY.

The Region Split: From DIM_COUNTRY, the hierarchy splits into two tables:

DIM_COUNTRY > DIM_ECONOMIC_REGIONS
DIM_COUNTRY > DIM_REGIONS